A timed example: the speed governor chain

19 October 2026. Companion to *complete.pdf*. Generated by `scripts/example_report.py`; every formula shown was checked by Z3 to be equivalent to what `scripts/delegate.py` computes. The instance is part of the one command `.venv/bin/python scripts/delegate.py`.

1 The instance

Three components in a chain. **Perception** reads the environment variable o (distance to an obstacle) and writes an estimate e. **Controller** reads e and writes a command u. **Actuator** reads u and writes the speed v. Horizon $H = 2$, so each variable has values at times $0, 1, 2$; all delays are 0. The query mentions only o and v; e and u are internal.

$$A_P = \Box_{[0,2]}\,(0 \le o \le 5), \qquad G_P = \Box_{[0,2]}\,(0 \le v \le 2) \;\wedge\; \Box_{[0,1]}\big(o \le 1 \to \Diamond_{[0,1]}\, v \le \tfrac{1}{4}\big) \;\wedge\; \Box_{[0,1]}\big(o \ge 3 \to \Diamond_{[0,1]}\, v \ge 1\big).$$

In words: the speed stays in $[0, 2]$; whenever the obstacle is close ($o \le 1$) the speed drops to at most $\frac{1}{4}$ within one step; whenever the way is clear ($o \ge 3$) the speed reaches at least 1 within one step. Unrolled, the two response conjuncts become, for $t = 0, 1$: $(o(t) \le 1 \to v(t) \le \frac{1}{4} \vee v(t{+}1) \le \frac{1}{4})$ and $(o(t) \ge 3 \to v(t) \ge 1 \vee v(t{+}1) \ge 1)$.

2 What the procedure does

Steps 1–3. All three components can influence v, so all are relevant. The only topological ordering is Perception, Controller, Actuator. It is nested: Perception sees o, which Controller does not, and has the one edge variable e to carry it; Controller sees e, which Actuator does not, and has u; neither e nor u occurs in the query; only Perception reads the environment. Forwarding formulas: $F_1 = \bigwedge_t e(t) = o(t)$ and $F_2 = \bigwedge_t u(t) = e(t)$. The only terminal output is v, so the sentence is

$$\Phi = \forall o(0)\, \exists v(0)\; \forall o(1)\, \exists v(1)\; \forall o(2)\, \exists v(2)\;\; (A_P \to G_P),$$

read: for every obstacle distance at time 0 there is a speed at time 0, such that for every distance at time 1 there is a speed at time 1, and so on, making the query true.

Steps 4–5. Quantifier elimination decides Φ: **true**. **Step 6–7.** The witnesses, read back through the forwarding (o arrives at the Actuator as u), give the seed delegation:

component	assumption	guarantee (seed)
Perception	true	$e(t) = o(t)$ for $t = 0, 1, 2$
Controller	true	$u(t) = e(t)$ for $t = 0, 1, 2$
Actuator	true	$v(0) = 1, \quad v(1) = \frac{1}{8}, \quad v(2) = \begin{cases} 1 & u(1) \ge 3 \\ \frac{1}{8} & \text{otherwise} \end{cases}$ (on $0 \le u(t) \le 5$)

The witness is lazy by construction: it starts fast ($v(0) = 1$ answers a clear road at time 0 at once), goes slow at time 1 ($\frac{1}{8}$ answers a close obstacle at time 0 or 1), and at time 2 goes fast again exactly when the command says the road was clear at time 1. Z3 confirms (V1), (V2), and that each component is receptive and causal.

Steps 8–9 (sweep). Each contract is replaced, in order, by the weakest one that still works with the others, then the assumptions are recomputed. The result, written out (equivalences checked by Z3):

component	assumption	guarantee (final)
Perception	A_P	$\big(A_P \wedge o(1) \ge 3\big) \;\to\; \big(\bigwedge_t 0 \le e(t) \le 5 \;\wedge\; e(1) \ge 3\big)$
Controller	true	$\big(\bigwedge_t 0 \le e(t) \le 5 \wedge e(1) \ge 3\big) \;\to\; \big(\bigwedge_t 0 \le u(t) \le 5 \wedge u(1) \ge 3\big)$
Actuator	true	$\bigwedge_t 0 \le v(t) \le 2 \;\wedge\; (v(0) \ge 1 \vee v(1) \ge 1) \;\wedge\; (v(0) \le \frac{1}{4} \vee v(1) \le \frac{1}{4}) \;\wedge\; (v(1) \le \frac{1}{4} \vee v(2) \le \frac{1}{4})$ $\wedge\, \big(\bigwedge_t 0 \le u(t) \le 5 \wedge u(1) \ge 3 \;\to\; v(1) \ge 1 \vee v(2) \ge 1\big)$

(V1), (V2), receptiveness and causality hold after the sweep (Z3).

3 Reading the result

The final contracts say precisely what each component must preserve, and nothing more.

- **Perception** must preserve exactly one feature: if the road is clear at time 1 ($o(1) \geq 3$), the estimate at time 1 must also say so ($e(1) \geq 3$) and stay in range. Nothing else about o has to reach the chain. The reason is in the Actuator's final contract: a close obstacle is handled without information, by being slow at time 1 and again at time 1 or 2, and a clear road at time 0 by being fast at time 0 or 1; the only case that needs information from the environment is the clear road at time 1, which must be answered at time 1 or 2 and cannot be answered blindly without breaking the slow-down obligation.
- **Controller** must pass that one feature on: estimate in range and $e(1) \geq 3$ implies command in range and $u(1) \geq 3$.
- **Actuator** must keep the speed in range, be fast at time 0 or 1, be slow at time 0 or 1 and at time 1 or 2, and, whenever the command certifies a clear road at time 1, be fast at time 1 or 2. Its final contract is weaker than the seed strategy: for instance $v(0) = \frac{1}{8}$, $v(1) = 1$ is also allowed.

The exact copies $e = o$, $u = e$ of the seed are gone; what remains on the edges is the one implication the query actually needs. This is the sweep doing its job: the forwarding seed only serves to find a solution, and quotient plus projection then remove everything the query does not require.

4 A refutation

Change the query to demand an immediate slow-down and a fast response exactly one step later: $G'_P = \square_{[0,2]}(0 \leq v \leq 2) \wedge \square_{[0,2]}(o \leq 1 \rightarrow v \leq \frac{1}{4}) \wedge \square_{[0,1]}(o \geq 3 \rightarrow \Diamond_{[1,1]}\, v \geq 1)$. The sentence Φ is **false**: the environment can present a clear road at time 0 and a close obstacle at time 1, which demands $v(1) \geq 1$ and $v(1) \leq \frac{1}{4}$. The procedure reports that no acceptable delegation exists, by the necessity theorem, before any contract is written.